

```
PS C:\Users\student\Desktop\IWN25CS212> python -u "c:\Users\student\Desktop\1BF24CS283\MANHATTAN DISTANCE.py"
A* SEARCH - 8 PUZZLE
```

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```

Initial State:

```
(2, 8, 3)
(1, 6, 4)
(7, 0, 5)
```

Goal State:

```
(1, 2, 3)
(8, 0, 4)
(7, 6, 5)
```

CASE 1: MISPLACED TILES

```
-----
```

Solution found!

Number of moves: 5

Step 0

```
(2, 8, 3)
(1, 6, 4)
(7, 0, 5)
```

Step 1

```
(2, 8, 3)
(1, 0, 4)
(7, 6, 5)
```

Step 2

```
(2, 0, 3)
(1, 8, 4)
(7, 6, 5)
```

(7, 6, 5)

Step 2

(2, 0, 3)

(1, 8, 4)

(7, 6, 5)

Step 3

(0, 2, 3)

(1, 8, 4)

(7, 6, 5)

Step 4

(1, 2, 3)

(0, 8, 4)

(7, 6, 5)

Step 5

(1, 2, 3)

(8, 0, 4)

(7, 6, 5)

CASE 2: MANHATTAN DISTANCE

-----

Solution found!

Number of moves: 5

Step 0

(2, 8, 3)

(1, 6, 4)

(7, 0, 5)

Step 1

(2, 8, 3)

Solution found!

Number of moves: 5

Step 0

(2, 8, 3)

(1, 6, 4)

(7, 0, 5)

Step 1

(2, 8, 3)

(1, 0, 4)

(7, 6, 5)

Step 2

(2, 0, 3)

(1, 8, 4)

(7, 6, 5)

Step 3

(0, 2, 3)

(1, 8, 4)

(7, 6, 5)

Step 4

(1, 2, 3)

(0, 8, 4)

(7, 6, 5)

Step 5

(1, 2, 3)

(8, 0, 4)

(7, 6, 5)

PS C:\Users\student\Desktop\IWN25CS212> 1BF24CS283 SHIWANI SINGH